Language Agent Tree Search

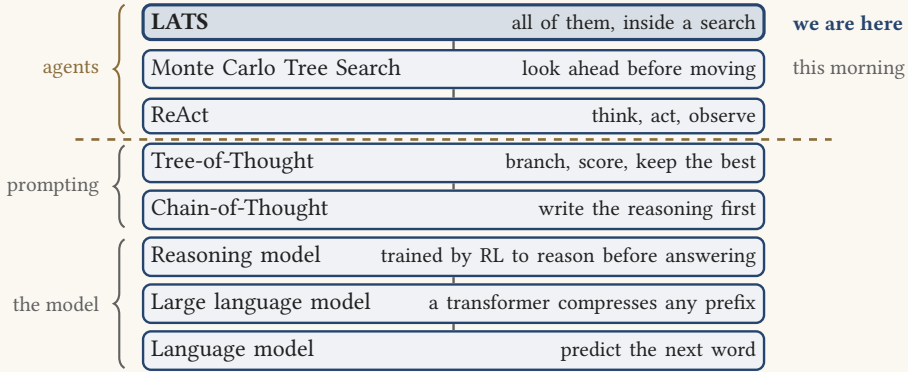
One frozen model as a planning agent

Carlos Cotrini

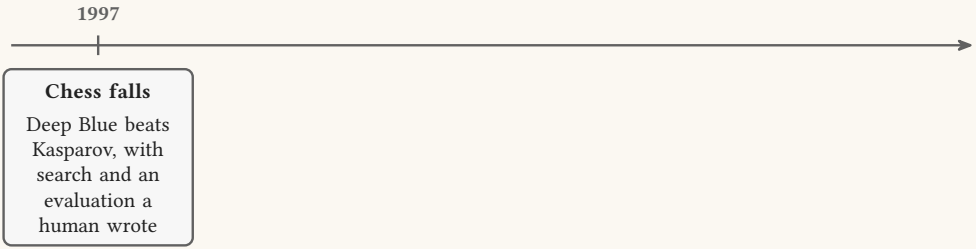
ETH Zurich

21 October 2026

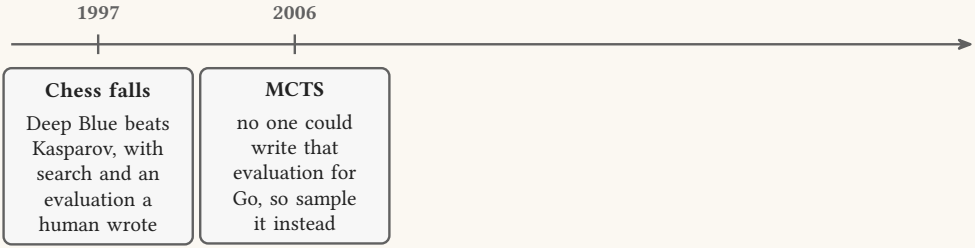
Where LATS sits



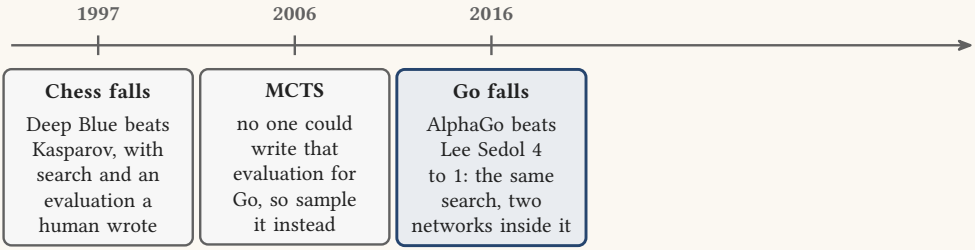
Monte Carlo Tree Search, before language models



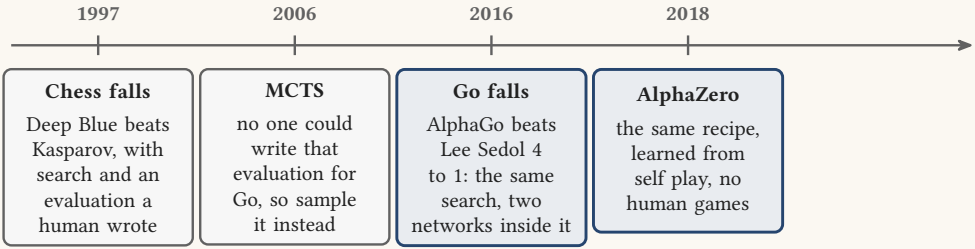
Monte Carlo Tree Search, before language models



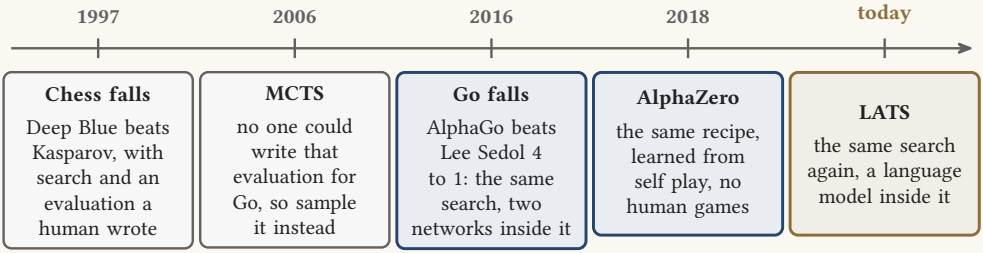
Monte Carlo Tree Search, before language models



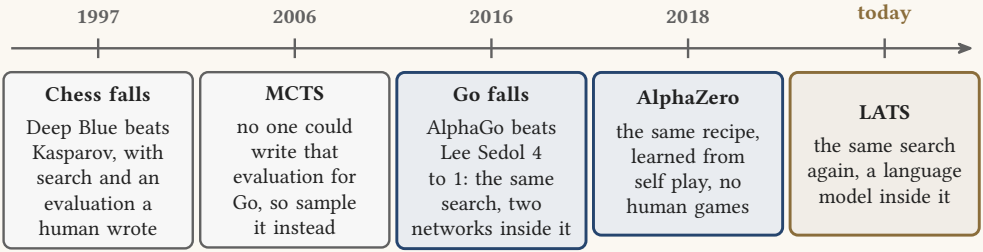
Monte Carlo Tree Search, before language models



Monte Carlo Tree Search, before language models



Monte Carlo Tree Search, before language models



The search itself barely changed across twenty years. What changed each time is what supplies the two things it needs: something to propose moves, and something to judge a position. LATS supplies both from one frozen model.

Coulom 2006; Kocsis and Szepesvari 2006; Silver et al., Nature 2016; Silver et al., Science 2018.

1

Section 1

The LATS method

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

The critic

what the model thinks this branch is worth

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

The critic
what the model thinks this branch is worth

+

The exploration bonus
how little we have tried it

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

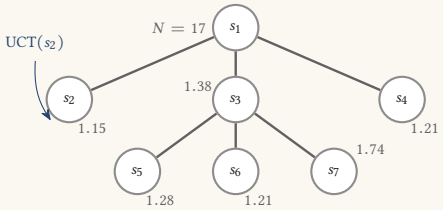
The critic
what the model thinks this branch is worth

+

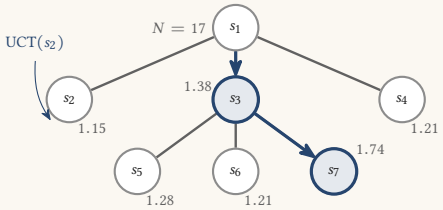
The exploration bonus
how little we have tried it

This morning's words for the same two terms: how good it looked, plus how little we know.

The same loop as this morning, with four edits

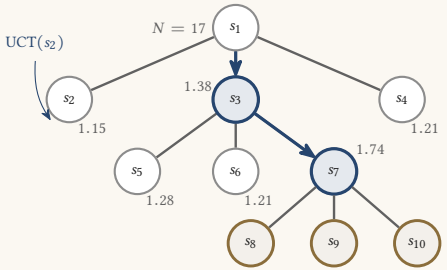


The same loop as this morning, with four edits



1 Selection by UCT

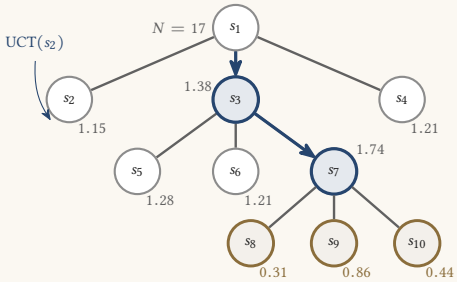
The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

The same loop as this morning, with four edits

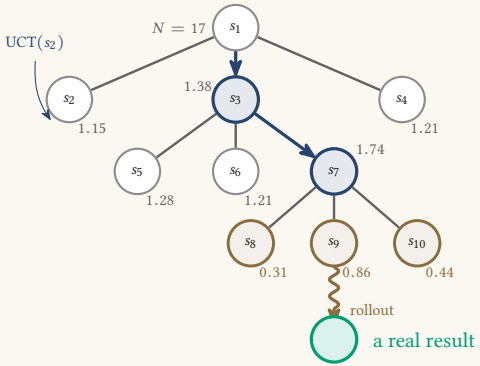


1 Selection by UCT

2 Expansion the *actor* proposes

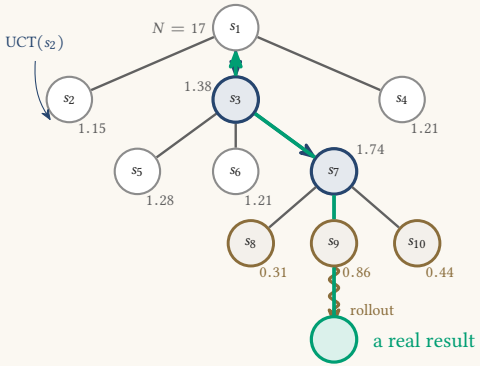
3 Evaluation the *critic* scores

The same loop as this morning, with four edits



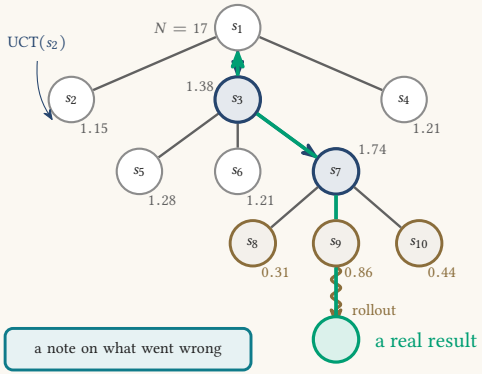
1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end

The same loop as this morning, with four edits



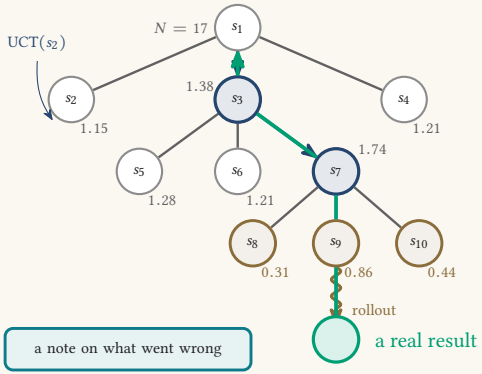
1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end
5 Backpropagation	the *real* result, up

The same loop as this morning, with four edits



1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end
5 Backpropagation	the *real* result, up
6 Reflection	the *feedback* model writes it

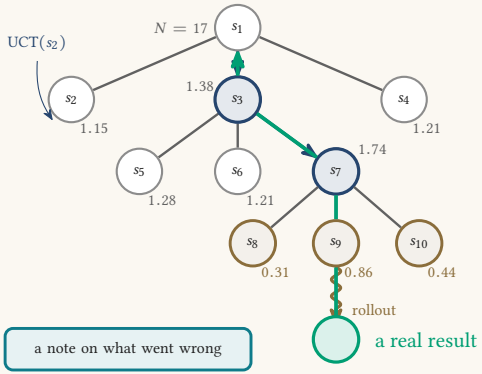
The same loop as this morning, with four edits



1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end
5 Backpropagation	the *real* result, up
6 Reflection	the *feedback* model writes it

grey: UCT, what selection compares.
amber: V from the critic, and all that is new.

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

5 Backpropagation the *real* result, up

6 Reflection the *feedback* model writes it

into the next prompt

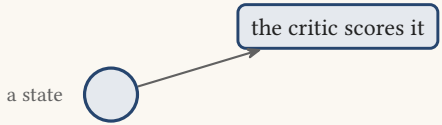
grey: UCT, what selection compares.
amber: V from the critic, and all that is new.

What the critic actually returns

a state 

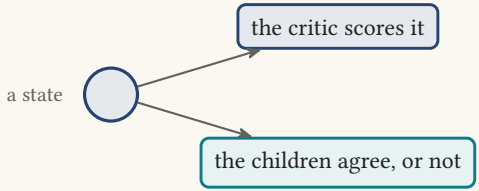
$$V(s) = \qquad +$$

What the critic actually returns



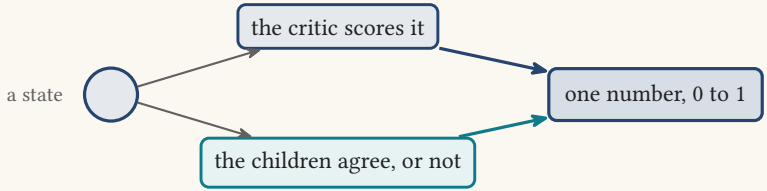
$$V(s) = \qquad +$$

What the critic actually returns



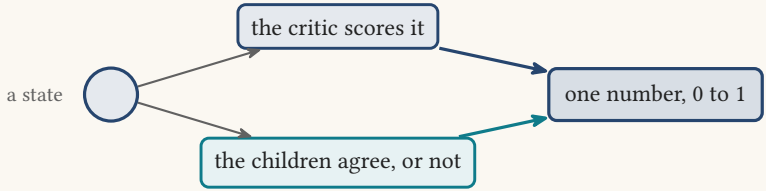
$$V(s) = \qquad\qquad +$$

What the critic actually returns



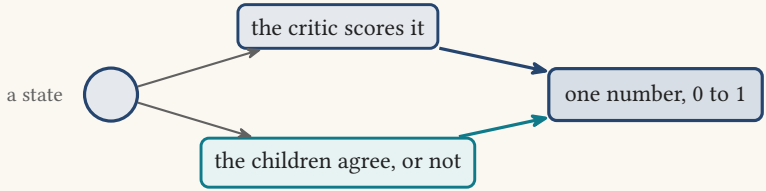
$$V(s) = \qquad +$$

What the critic actually returns



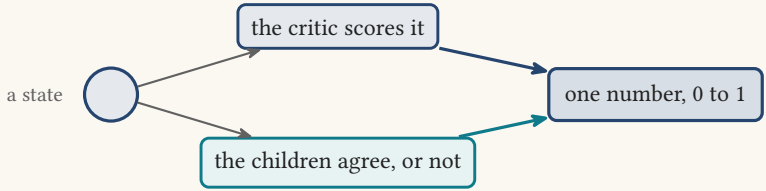
$$V(s) = \lambda\,\mathrm{LM}(s) +$$

What the critic actually returns



$$V(s) \;=\; \lambda\,\mathrm{LM}(s) \;+\; (1-\lambda)\,\mathrm{SC}(s)$$

What the critic actually returns



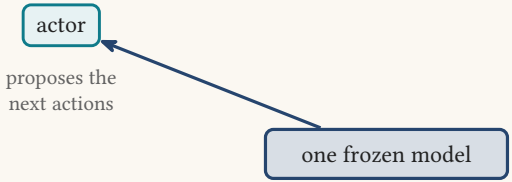
$$V(s) \;=\; \lambda\,\mathrm{LM}(s) \;+\; (1-\lambda)\,\mathrm{SC}(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

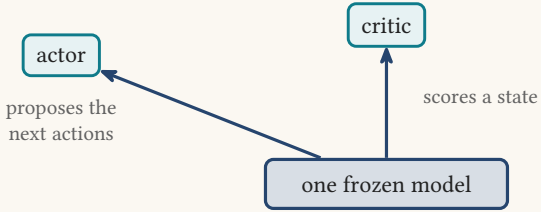
Three jobs, one set of weights

one frozen model

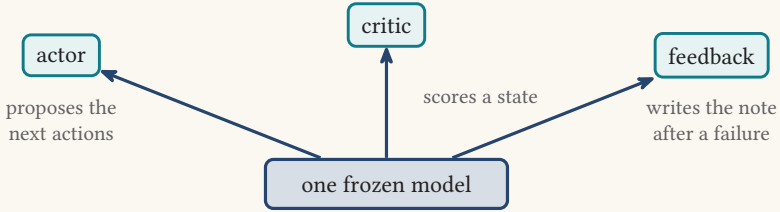
Three jobs, one set of weights



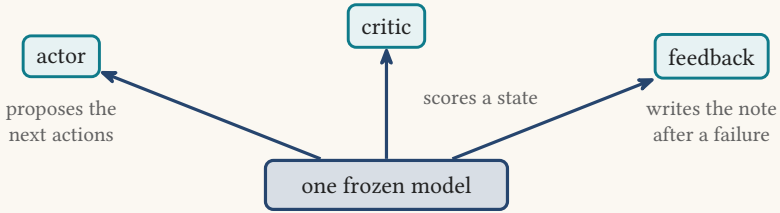
Three jobs, one set of weights



Three jobs, one set of weights



Three jobs, one set of weights



same weights every time. Only the prompt changes.